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Internship Assessment Task

Hand Detection and Tracking System

Duration: 2 Days • Dheera AI

Overview

Build a simple hand detection system that can process webcam video and display results in a web interface. This quick task will assess your ability to integrate computer vision with web development while effectively using AI coding assistants.

Objective

Create a working prototype that detects hands in real-time video and displays the results. Focus on getting a functional MVP rather than perfection.

Core Requirements

Must-Have Features

- Detect hands in webcam video using any computer vision library
- Display bounding boxes or landmarks on detected hands
- Show live video feed with detection overlays in a web browser
- Basic UI with start/stop controls

Bonus Features (Optional)

If you finish early or want to showcase additional skills:

• **Gesture recognition:** Detect specific hand gestures (thumbs up, peace sign, etc.) • **Hand tracking:** Track individual hands across frames with unique IDs • **Visualization options:** Allow users to toggle between skeleton view, bounding boxes, or heatmaps
• **Recording capability:** Let users record and download processed video • **Multi-hand support:** Handle detection of multiple hands simultaneously

Technical Guidelines

Technology Choices

You're free to choose your tech stack, but here are some suggestions:

• **Computer Vision:** MediaPipe, OpenCV, TensorFlow.js, or any CV library • **Backend:** Python (Flask/FastAPI), Node.js, or any server framework • **Frontend:** React, Vue, vanilla JavaScript, or your preferred framework

Use AI Coding Tools

We encourage you to leverage AI coding assistants throughout this task:

- Use tools like GitHub Copilot, Claude code, Antigravity, or others to accelerate development

• Ask AI tools for boilerplate code, debugging help, and implementation suggestions

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• Document which tools you used in your README • Show that you can validate and debug AI-generated code effectively

Deliverables

1. **Working Code**

Push to GitHub (public or private with access granted to 'yash-dheera')

2. **Simple README**

Include:

• How to run the application (setup instructions)
• Dependencies/requirements
• Which AI tools you used and how
• Any challenges you faced

3. **Demo Video OR Screenshots**

1-2 minute video showing it working, OR 2-3 clear screenshots of the interface with detection running

Evaluation Criteria

Your submission will be evaluated on:

• **Does it work? (50%):** Can we run it and see hand detection happening? • **Code clarity (20%):** Is the code reasonably clean and understandable? • **User Experience (15%):** Interface design and interaction flow
• **Setup ease (15%):** Can we follow your README and get it running? • **Problem-Solving (10%):** How you tackled challenges and made trade-offs

Bonus points for creativity, performance optimization, and thoughtful use of AI tools!

Submission Instructions

Send an email to **[tech@dheera.co]** with:

• GitHub repository link
• Demo video link

Subject Line: "Hand Detection Task Submission - [Your Name]"

Tips for Success

• **Start simple:** Get basic functionality working first, then add features • **Use AI tools liberally:** Ask for example code, explanations, and debugging help • **Focus on working > perfect:** A simple working demo beats incomplete fancy features • **Ask questions:** If requirements are unclear, reach out for clarification • **Focus on core features:** A working MVP is better than an incomplete feature-rich version